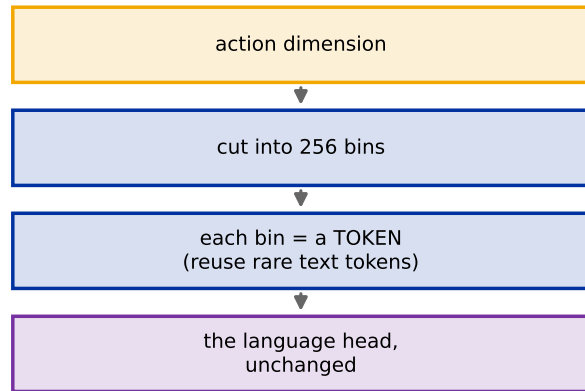


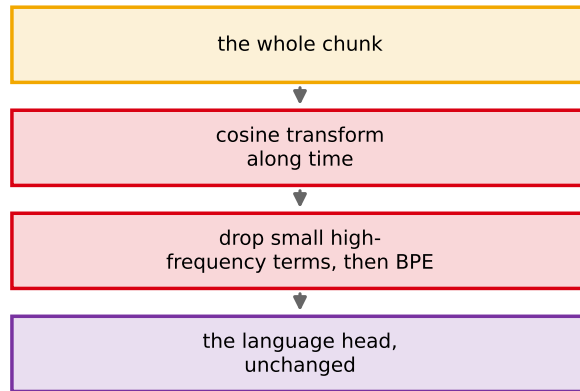
The same fork as chapter 10 and chapter 13: discretize and reuse the language head, or model the numbers directly.

1. discrete bins
RT-2, OpenVLA



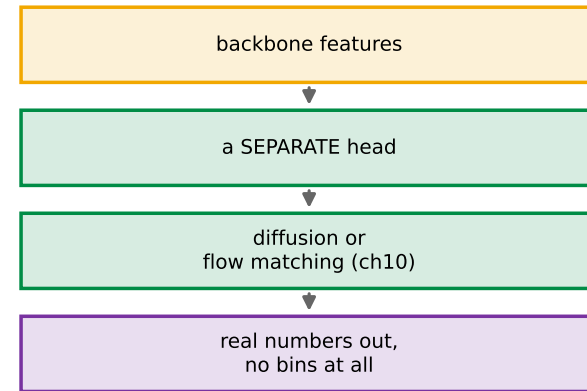
*7 dims x 50 steps
= 350 tokens. Slow.*

2. compressed tokens
FAST



*~10x fewer tokens.
The JPEG trick.*

3. continuous head
Octo, pi-0, GR00T N1



*Can represent BOTH
ways around the mug.*